

Modernizing Operational Data

Salesforce integration and automation across lead capture, registrations, product commerce, and business reporting.

The organization did not have a data shortage. It had a coordination problem: useful information existed across CRM, ecommerce, web forms, registration tools, and marketing channels, but not in one dependable operating model.

CAPTURE Leads, registrations, products, orders	NORMALIZE Formats, arrays, price records, identifiers	ORCHESTRATE Routing, sync, validation, follow-up	INTERPRET Attribution, demand, revenue, operations
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Sanitization note

Organization names, URLs, record IDs, credentials, employee names, course names, and proprietary configuration details have been removed or generalized. The architecture, design decisions, and operational lessons are preserved.

The work moved the business from disconnected transactions toward a governed data foundation - one where operational systems could exchange information reliably and leadership could begin asking better questions of the data already being collected.

Several systems worked. The business view did not.

CRM, ecommerce, course registrations, website inquiries, and marketing channels each handled their immediate job. The friction appeared when a person needed to trace an outcome across more than one system.

WEBSITE & CAMPAIGNS

High-intent engagement

Questions, consultation requests, private-training interest, tagged links, event traffic, and QR-code responses arrived with useful context that was not always preserved downstream.

CRM

Relationship records

Leads, contacts, accounts, courses, products, and registrations existed, but field definitions and ownership rules had to be made consistent before the data could support automation or reporting.

COMMERCE

Products, pricing & orders

Product records and multiple price books lived in Salesforce while storefront execution, order data, shipping behavior, and categories lived in WooCommerce.

REGISTRATION OPERATIONS

People, courses & purchases

Participation data needed to preserve the links among people, companies, courses, attendance, payment, and related product purchases.

The design shift

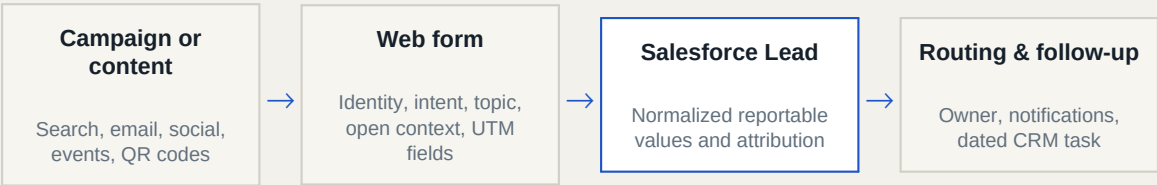
The original instinct was to move records between tools. The stronger architecture focused on a trustworthy analytical spine: stable identifiers, normalized fields, explicit ownership, validation, and repeatable synchronization.

Questions the new foundation was designed to support:

- Which campaigns or content create meaningful leads and downstream opportunities?
- Which product families, pricing structures, and customer segments drive purchasing behavior?
- Which courses, accounts, contacts, and purchases belong to the same customer journey?
- Where are missing prices, unmatched records, incomplete fulfillment fields, or broken relationships creating risk?

Turn digital interest into a structured, accountable sales process.

A reusable website form captured just enough information to qualify the inquiry while preserving the visitor experience. Behind the scenes, form data was transformed into CRM-compatible values, hidden campaign parameters were retained, and Salesforce automation took responsibility for routing and follow-up.



Data manipulation mattered. Website checkbox lists arrived in a visitor-friendly format while Salesforce multi-select fields required a different delimiter. A custom transformation converted each selection before transmission so separate choices stayed distinct and reportable.

Automation created accountability. Record-triggered logic evaluated intent in a controlled order, assigned one owner, notified additional staff when relevant, and created a visible next action instead of relying on shared inboxes.

Identity	Name, company, email, phone, title
Intent	Training, consulting, credentials, membership, support
Context	Topic interests, notes, submitted question
Attribution	Source, medium, campaign

Consistency

Different entry points feed the same structured lead framework.

Context

Staff see why the prospect engaged before the first conversation.

Accountability

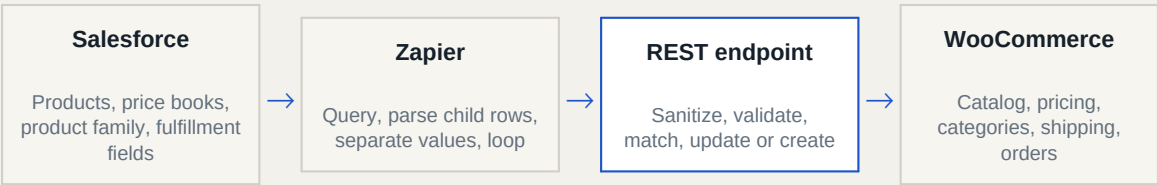
Ownership, notifications, and tasks make follow-up visible.

Attribution

Marketing activity remains attached to CRM outcomes instead of disappearing at form submission.

Use automation as a controlled data contract between CRM and ecommerce.

Salesforce remained the source for core product and price-book data. Zapier queried and transformed records, while a custom WordPress REST endpoint acted as the controlled write interface into WooCommerce.



Price-book handling. Multiple child price records could not be treated as one value. The workflow explicitly identified each price-book entry and emitted separate standard and member prices before posting. A fallback preserved catalog completeness when the member price was absent.

Stable matching. Product names were rejected as keys. SKU remained useful within an environment, but cross-environment reconciliation required a stable business identifier such as Product Code or another dedicated external key.

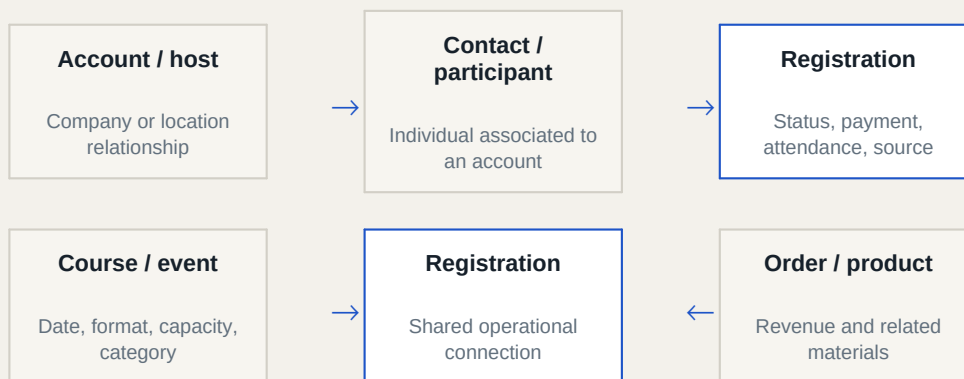
Endpoint validation. The endpoint isolated WooCommerce implementation details from the automation layer and handled required fields, sanitization, create-versus-update behavior, categories, dimensions, virtual status, pricing metadata, and traceability fields.

Fulfillment-ready data. Weight, dimensions, virtual status, categories, and other operational attributes were treated as data-quality requirements, not optional storefront decoration.

01	02	03	04	05
Define the source of truth for each field.	Separate identifiers from display labels.	Transform child records before posting.	Validate at the endpoint.	Make missing data reportable.

Treat a registration as a relationship-rich record, not an isolated transaction.

Course activity only becomes useful for analysis when the registration preserves the links among the participant, organization, course, payment, order, and related materials. The model therefore centered relationships and stable identifiers rather than relying on form labels or names.



Variable attendee counts. A looping workflow supported submissions with one or many attendees while creating the appropriate participant and registration records instead of flattening every attendee into one unusable payload.

Account and contact integrity. Company data was checked before creating new account records, then participants were associated with the correct account to reduce duplicate or split relationship histories.

Commerce context. Order billing information and product identifiers were carried into the workflow so registration activity could remain connected to the purchase that produced it.

Separate processes, shared model. Registration and ticket automation remained distinct from product synchronization, but both were designed to converge on governed identifiers and relationships inside the broader operating model.

Decision-ready questions enabled by the model

Which course families are growing? Which accounts return? Where are cancellations concentrated? Which products are purchased with a course? How do member status, price, registration history, and order behavior interact?

The highest-value output was a business that could begin trusting its own operational data.

The individual automations solved immediate problems, but together they created something more durable: a governed foundation for accumulation, manipulation, and interpretation of data across the customer lifecycle.

ACCUMULATION

Capture useful context early

Intent, topic, attribution, pricing, product attributes, registration status, and relationship keys are collected before they disappear into disconnected transactions.

MANIPULATION

Make systems compatible

Arrays, multi-select values, child records, prices, and identifiers are transformed into explicit fields that downstream systems can validate and report.

OPERATIONS

Automate accountable handoffs

Ownership, alerts, tasks, create/update controls, and endpoint validation reduce the amount of judgment hidden inside manual reconciliation.

INTERPRETATION

Build toward decision-ready reporting

Marketing activity, customer relationships, participation, products, orders, pricing, and fulfillment data can be connected without rebuilding the operating layer for every new question.

Implementation lessons carried forward:

- Model the data before mapping the automation. Field types, valid values, and ownership determine reporting quality later.
- Names are labels, not keys. Stable identifiers are the difference between synchronization and duplicate creation.
- Transform before transmitting. Multi-selects, arrays, line items, and child price records need explicit handling.
- Validate at system boundaries. API endpoints should sanitize, enforce required fields, and make create-versus-update behavior predictable.
- Treat data-quality exceptions as data. Missing dimensions, prices, categories, relationships, and attribution fields should become visible before they distort decisions.

Final perspective

Useful business intelligence begins with operational discipline. Once leads, products, prices, registrations, contacts, accounts, and orders share reliable identifiers and definitions, the organization can move from reactive reconciliation toward repeatable analysis and more confident decisions.